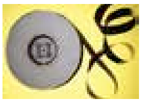

Magnetic tape memory
The same old tech used in audio casettes is still used for data storage, even by tech giants. **https://dgit.in/jul20-91**

History of HDDs
The first hard disk drives were massive enough to fill up a room, and could only hold 5MB of data. **https://dgit.in/jul20-92**

consisted of eight individual memory boards, each packing 256 KB of RAM chips. The Bulk Core could provide a massive 2 MV of storage capacity and its access times ranged from 0.75 milliseconds to 2 milliseconds. In contrast, today we have SSDs with 0.06ms access times. In 1988, an Alabama-based PC vendor called Digipro unveiled a prototype of the first SSD that used flash memory, capitalising on Intel's NOR flash memory chips that were released earlier that year. Named Flashdisk, the Digipro SSD could hold up to 16 MB of data. These were, however, exorbitantly priced and could cost tens of thousands of dollars per drive. In 2003, however, Transcend introduced a line of affordable SSDs that emulated Parallel ATA IDE hard drives. However, each module was smaller than a traditional PATA hard drive. The drives shipped in 16 MB to 512 MB capacities at the time and the later years saw larger capacities follow. In 2006, Samsung also

The early DVD players we used to proudly display in our living rooms

released its first mass-market flash SSD, and then SanDisk followed suit too. Flash SSDs in 2006 were capable of more rewrites than flash media cards, which brought them closer to replacing hard drives for everyday use. The SSD market began growing exponentially and continues to flourish and grow even today.

The next invention in the evolution of data storage was the Compact Disc, more commonly referred to by its abbreviation, CD. The compact disc was a form of optical storage that was first developed in 1982 through a conjoint effort by two companies – Sony and Philips. Although the CD was only 12 centimeters in diameter when it was first developed, it could manage to store more data than that of a personal PC's hard drive! CD drives read data stored on discs by shining a focused laser beam on the surface of the disc. CDs eventually replaced vinyl records and cassette tapes in the music industry. At this time, CDs were thought of as an evolution of the gramophone. The applications in generalised data storage came into the limelight soon after due to its capacity and portability and this spawned data storage options such as CD-ROM, CD-R, CD-RW, VCD, SVCD, PhotoCD, DVDs and Blu-Ray.

The first Digital Video Disk (DVD) had a whopping 1.46 GB of storage capacity which was enough to hold an entire short movie. Some manufacturers also began making the DVDs dual-sided to hold more data. This form of storage was also developed by Sony and Philips in 1995, along with an array of other technology companies. The DVD stores data using the same optical functions as the CD, only with higher storage capacities. The inception of DVDs shook up the film industry and phased out the widely-used VHS. However, the DVD itself was soon after eclipsed by the tremendous capabilities of the Blu-Ray disc in 2003. The Blu-Ray disc stored up to 25 GB of HD video at 1080p. Today, Sony has cranked up this number to 3.3 TB of data storage! It dwarfed the 480p resolution and the meagre storage capacity of DVDs.

THE AGE OF PORTABILITY

USB Flash Drives, a 'plug-and-play' storage device, was developed by SanDisk (then M-Systems) in 1999. They called the device DiskOnKey and it had a capacity of 8 MB. The device was brought to market in 2000. USB Flash Drives are lightweight, ultra-portable and are also removable and rewritable storage devices. Today, the maximum capacity of a flash drive is a massive 2 TB! It is also colloquially known as pen drive, thumb drive, jump drive, or a memory stick.

The miniscule form factor and massive storage capacity of SD cards would have brought a tear to the eyes of early data storage developers

Another ultra-portable storage device that was developed in 1999 by SanDisk, Panasonic and Toshiba conjointly is the SD card. The technology was built on previous iterations of it including the MultiMediaCard (MMC). SD cards use flash memory and store data in cells made of floating-gate transistors. These are popularly used in digital cameras and smartphones. Through the years, mini and micro versions of the SD card were also released. The first SD card held about 64 MB of data while the highest capacity SD card today can hold 1 TB of data.

DATA TAKES FLIGHT

While John McCarthy first introduced the concept of cloud computing in 1961, it wasn't until Amazon unveiled AWS in 2006 that the idea of cloud really took off. Even though AT&T did offer an all web-based data storage solution called PersonaLink Services way back in 1994, Amazon's AWS truly started the trend of cloud data storage. Companies moved to cloud-based solutions and slowly, but surely, consumers started adopting cloud services to store their ever-engorging data as well. Now, your data has a limitless home and your storage capacity depends not on the capabilities of the device, but the plan you pay for. You get remote access to databases, meaning your data is accessible anytime, anywhere, as long as you have an internet connection. Moreover, cloud storage services are bound to get cheaper over time as the technology improves, making it a truly accessible mass data storage solution.